

```

1: ////////////////////////////////// Test 2, 2023/2024 (Sem 1) //////////////////////////////////
2: ///// QUESTION 1
3: ///// Name: Nurul Ika Syafiny Bt Azhar
4: ///// Matric No: A23CS0164
5: ///// Date / Day: 4/1/2024 Thursday
6: //////////////////////////////////
7:
8: #include <iostream>
9: #include<iomanip>
10: using namespace std;
11:
12: // function prototypes
13: float price_original();
14: char choose_option();
15: void price_afterTax(float * , char);
16:
17: // start main function
18: int main()
19: {
20:     float price = 0; // should > 0
21:     char option; // A -> Encyclopedia
22:                // B -> Textbook
23:                // C -> Novel
24:                // D -> Magazine
25:     option = choose_option();
26:     price = price_original();
27:     cout << "Original book price before tax = RM " << price << "\n\n";
28:     price_afterTax(&price, option);
29:     cout << "Total book price after tax = RM " <<fixed<<setprecision(2)<<price << '\n';
30:     return 0;
31: }
32:
33: // start new user-defined functions
34: float price_original()

```

```

35: {
36:     float bookPrice = 0;
37:     while (bookPrice <= 0)
38:     {
39:         cout << "Please insert the book price (RM): ";
40:         cin >> bookPrice;
41:     }
42:     return bookPrice;
43: }
44:
45: char choose_option()
46: {
47:     char category;
48:     do
49:     {
50:         cout << "Please choose the book category: \n";
51:         cout << "A -> Encyclopedia\n";
52:         cout << "B -> Textbook\n";
53:         cout << "C -> Novel\n";
54:         cout << "D -> Magazine\n";
55:         cout << "Book category (A,B,C,D): ";
56:         cin >> category;
57:         cout << "\n";
58:     } while ((category < 'A') || (category > 'D'));
59:     return category;
60: }
61:
62: // tax is based on book categories as follows:
63: // A -> Encyclopedia => 3%
64: // B -> Textbook => 2%
65: // C -> Novel => 1%
66: // D -> Magazine => 0.5%
67: void price_afterTax(float *bookPrice, char category)
68: {

```

```
69:     float tax;
70:     switch (category)
71:     {
72:         case 'A':
73:             tax = *bookPrice * (3.0 / 100);
74:             break;
75:         case 'B':
76:             tax = *bookPrice * (2.0 / 100);
77:             break;
78:         case 'C':
79:             tax = *bookPrice * (1.0 / 100);
80:             break;
81:         case 'D':
82:             tax = *bookPrice * (0.5 / 100);
83:             break;
84:     }
85:     cout << "Tax price = RM " << fixed << setprecision(2) << tax << "\n";
86:     *bookPrice += tax;
87: }
88:
```